Week 1 | TA Session - 1

Review of Linear Algebra

Pulkit Mangal

1. Notations:

1.1. Vectors:

$$\mathbf{v} \in \mathbb{R}^d$$

column vector

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_d \end{bmatrix}$$

row vector

$$\mathbf{v}^T = \begin{bmatrix} v_1 & \cdots & v_d \end{bmatrix}$$

1.2. Matrix:

$$\mathbf{A} \in \mathbb{M}^{m \times n}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

- m rows and n columns.
- column vectors are in dimension $\mathbb{R}^m$
- row vectors are in dimension $\mathbb{R}^n$

2. Vector operations

2.1. Inner Product(Dot Product):

$$\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix}$$

Dot product on the above vectors is defined as :

$$\begin{bmatrix} x_1 & \cdots & x_d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{bmatrix} = \sum_{i=1}^{d} x_i y_i = \mathbf{x}^T \mathbf{y}$$

Note: For dot product, the vectors on which we are performing has to be of same dimensions.

2.2. Outer Product:

$$\mathbf{x} \in \mathbb{R}^d, \mathbf{y} \in \mathbb{R}^p$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix}$$

The **outer product** $\mathbf{A} = \mathbf{x} \otimes \mathbf{y}$ is a matrix defined by:

$$\mathbf{A} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \begin{bmatrix} y_1 & \cdots & y_p \end{bmatrix} = \mathbf{x}\mathbf{y}^T = \sum_{i=1}^{m} \sum_{j=1}^{p} x_i y_j$$

$$= \begin{bmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_p \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_p \\ \vdots & \vdots & \ddots & \vdots \\ x_d y_1 & x_d y_2 & \cdots & x_d y_p \end{bmatrix}$$

Note: For outer product, the dimensions of the vectors on which we are performing need not be same.

Important: The matrix obtained using outer product between two vectors is a **rank-1** matrix. As the columns of the resulting matrix are just scalar multiples of the first column.

Inner Product/Dot product	**Outer Product**
The dimensions has to be same	The dimensions need not be same
Output is a scalar	Output is a matrix
It is symmetric	It need not be symmetric.

3. Matrix operations

$$\mathbf{A} \in \mathbb{M}^{m\times n}, \mathbf{B} \in \mathbb{M}^{n\times p}\ \mathbf{x} \in \mathbb{R}^n$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{np} \end{bmatrix},\ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

There are two ways or forms in which one can do the operation $\mathbf{Ax}$

3.1. Matrix-Vector operations

3.1.1. Traditional approach/Dot product form

$$\mathbf{A} = \begin{bmatrix} - & R_1 & - \\ & \vdots & \\ - & R_m & - \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Here we will take each row of the matrix, which will be a vector in $\mathbb{R}^n$ and do the dot product with vector $\mathbf{x}$.

$$\mathbf{Ax} = \begin{bmatrix} R_1 . \mathbf{x} \\ \vdots \\ R_m . \mathbf{x} \end{bmatrix}$$

3.1.2. Linear combination of columns form

$$\mathbf{A} = \begin{bmatrix} | & & | \\ C_1 & \cdots & C_n \\ | & & | \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The way to do this is we take each element of vector $\mathbf{x}$ and scale the respective column of matrix $\mathbf{A}$ by that element and sum all the scale element.

$$\mathbf{Ax} = \begin{bmatrix} | \\ x_1 C_1 + \cdots + x_n C_n \\ | \end{bmatrix}$$

Important: The above case is equivalent to a linear combination of the columns of the matrix, where the multipliers come from the vector.

In both the case, we will get the same vector in $\mathbb{R}^{\mathsf{m}}$.

3.1.3. Vector-Matrix operation

$$\mathbf{B} = \begin{bmatrix} - & R_1 & - \\ & \vdots & \\ - & R_n & - \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Beside going with traditional way, we can write product $\mathbf{x}^T\mathbf{B}$ as

$$\mathbf{x}^T\mathbf{B} = x_1 . R_1 + \cdots x_n . R_n$$

Here it will be a row vector in $\mathbb{R}^p$

3.2. Matrix-Matrix operations

3.2.1. Traditional approach

$$\mathbf{A} = \begin{bmatrix} - & R_1 & - \\ & \vdots & \\ - & R_m & - \end{bmatrix}, \ \mathbf{B} = \begin{bmatrix} | & & | \\ C_1 & \cdots & C_p \\ | & & | \end{bmatrix}$$

The product $\mathbf{AB}$ can be seen as dot product between rows of $\mathbf{A}$ and $\mathbf{B}$. Note that each row vector in the matrix $\mathbf{A}$ would be transpose of column vectors.

$$\mathbf{AB} = \begin{bmatrix} R_1 . C_1 & \cdots & R_1 . C_p \\ \vdots & \ddots & \vdots \\ R_m . C_1 & \cdots & R_m . C_p \end{bmatrix}$$

3.2.2. Second Way (Outer product form)

$$\mathbf{A} = \begin{bmatrix} | & & | \\ C_1 & \cdots & C_n \\ | & & | \end{bmatrix}, \mathbf{B} = \begin{bmatrix} - & R_1 & - \\ & \vdots & \\ - & R_n & - \end{bmatrix}$$

Another way to see the product $\mathbf{AB}$ is the sum of the matrices obtained using outer product on rows and columns

$$\mathbf{AB} = C_1 R_1 + \cdots C_n R_n$$

Notice that any pair $C_i R_i$ will give a $m \times p$ matrix.

3.2.3. Third Way (Matrix-Vector form)

$$\mathbf{B} = \begin{bmatrix} | & & | \\ C_1 & \cdots & C_p \\ | & & | \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} | & & | \\ \mathbf{A}C_1 & \cdots & \mathbf{A}C_p \\ | & & | \end{bmatrix}$$

In this form we can see that in the resultant matrix, any i^{th} column can be represented as $\mathbf{A}C_i$ where C_i is the ith column of matrix $\mathbf{B}$.

3.2.4. Fourth Way (Vector-Matrix form)

$$\mathbf{A} = \begin{bmatrix} - & R_1 & - \\ & \vdots & \\ - & R_m & - \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} - & R_1 B & - \\ & \vdots & \\ - & R_m B & - \end{bmatrix}$$

In this form we can see that in the resultant matrix, any i^{th} row can be represented as $R_i\mathbf{B}$ where R_i is the ith row of matrix $\mathbf{A}$.

4. Span & Basis

The span of a set S (of vectors) is defined as the set of all finite linear combinations of elements (vectors) of S and denoted by $\mathrm{Span(S)}$

$$\mathrm{Span}(S) = \left\{ \sum_{i=1}^{n} a_i v_i \in V \mid \ a_i \in \mathbb{R}, \ v_i \in S \right\}$$

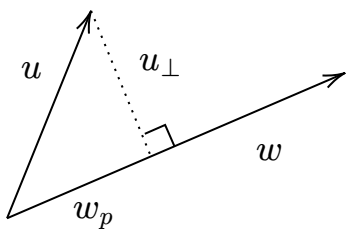
Let V be a vector space. A set $S \subseteq V$ is spanning set for V if $\mathrm{Span}(S) = V$.

The following conditions are equivalent for a set to be a **basis** of a vector space:

- **Maximal linearly independent set** : A linearly independent set such that if any vector is appended to it, the new set is no longer linearly independent.

- **Minimal spanning set** : A spanning set such that if any vector is removed from the set, then the set will not be a spanning set.

5. Projections



$$w_p = cw$$

$$u = u_\perp + w_p$$

Vector Projection

$$w_p = \frac{u^T w}{w^T w}\, w$$

Scalar Projection/Length of Projected Vector

$$\|w_p\| = \sqrt{\left(\frac{u^T w}{w^T w}\, w\right)^T \left(\frac{u^T w}{w^T w}\, w\right)}$$

Extending the above idea: The projection of a vector u onto a subspace W having orthogonal *basis* $\{w_1, w_2, \cdots w_k\}$ is given by

$$u_p = \sum_{i=1}^{k} \frac{u^T w_i}{w_i^T w_i}$$

Click here for visual animation.

6. Orthogonal Basis and Matrix

Two non-zero vectors are called to be orthogonal if the inner product (dot product) between them is 0. In the scenario of dot product as inner product, the angle between the vectors will be $90°$.

$$\cos\theta = \frac{u^T v}{\|u\|.\,\|v\|}$$

$$\text{if } u^T v = 0$$

$$\cos\theta = 0 \Longrightarrow \theta = 90°$$

- A set of vectors is called orthogonal set if each vector in the set is orthogonal to every other vector.
- The set of orthogonal vectors is linearly independent. Hence it form a basis of the subspace spanned by vectors.
- We can convert orthogonal vectors into orthonormal vectors by just dividing each vector by its norm.
- A $n \times n$ matrix Q is called orthogonal matrix if

$$Q^T Q = QQ^T = I_{n\times n}$$

- Orthogonal matrix preserve length and angle. That is $\|Qu\| = \|u\|$ for any vector u in $\mathbb{R}^n$.

7. Eigen-pair

Recall Linear Transformation: A function $T : V \rightarrow W$ is called linear transformation if it satisfies the following criteria -

$$T(v + \alpha w) = T(v) + \alpha . T(w)$$

where $v, w \in V$ and $\alpha \in \mathbb{R}$.

7.1. Eigen-vector

A non-zero vector is called eigen-vector for a matrix A if its direction (spanning set) does not change after multiplying with matrix A.

7.2. Eigen-value

The only change that will happen to vector is that either it will *shrink* or *expand*, the amount by which the vector will *shrink* or *expand* is called the eigen-value.

$$Av = \lambda v$$

8. Reference & Credits

- The content present in the slides have taken reference from the set of lectures recorded by Prof. Arun Rajkumar and the sessions by BS degree Instructor Karthik Thiagarajan, Vishal and Nitin for the course MLT. This would be the case for any future slides also.
- The content present in the slides can only be seen as an another way to represent the same thing taught by above mentioned authorities. At the same time, these slides are not meant to be a replacement for the lectures or live sessions conducted by instructors.
- The slides are prepared using mathcha.io.

Week-1 | Summary

Karthik Thiagarajan

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1. Misc

1.1. Notation

Scalars:

$$x_1, x_2, y_1, y_2, z_2, z_2, a, b, \alpha, \beta$$

Column vector:

$$\mathbf{x} \in \mathbb{R}^d$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

Row vector:

$$\mathbf{x} \in \mathbb{R}^d$$

$$\mathbf{x}^T = \begin{bmatrix} x_1 & \cdots & x_d \end{bmatrix}$$

Matrix:

$$\mathbf{X} \in \mathbb{R}^{d \times n}$$

1.2. Data-matrix

$$\mathbf{X} \in \mathbb{R}^{d \times n}$$

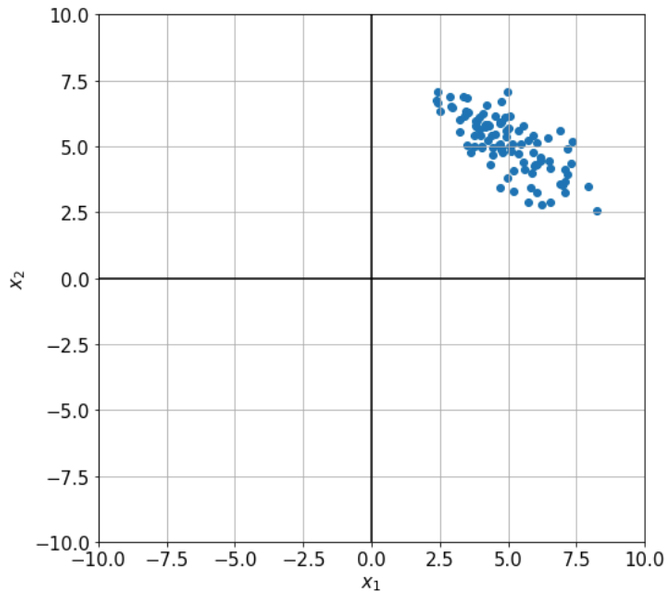
- $d \rightarrow$ number of features
- $n \rightarrow$ number of data-points

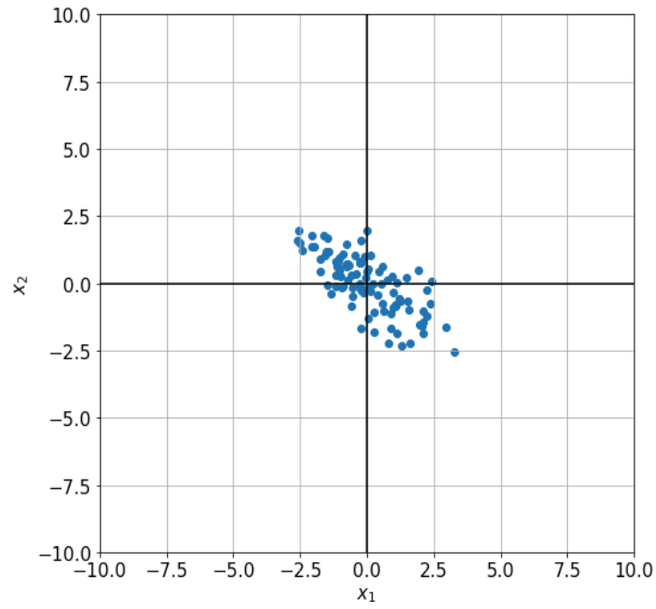
$$\mathbf{X} = \begin{bmatrix} | & & | \\ \mathbf{x}_1 & \cdots & \mathbf{x}_n \\ | & & | \end{bmatrix}$$

1.3. Data-point

$$\mathbf{x}_i \in \mathbb{R}^d$$

2. Centering the dataset





$$\overline{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$$

If the dataset is already centered, $\overline{\mathbf{x}} = \mathbf{0}$. If $\overline{x} \neq \mathbf{0}$, do the following:

$$\mathbf{x}_i' = \mathbf{x}_i - \overline{\mathbf{x}}$$

$$\mathbf{X}_c = \begin{bmatrix} | & & | \\ \mathbf{x}_1' & \cdots & \mathbf{x}_n' \\ | & & | \end{bmatrix}$$

$\mathbf{X}_c$ is the centered data-matrix.

Remark: From now we will work only with the centered data-matrix and will be calling it $\mathbf{X}$ (the subscript c will be dropped)

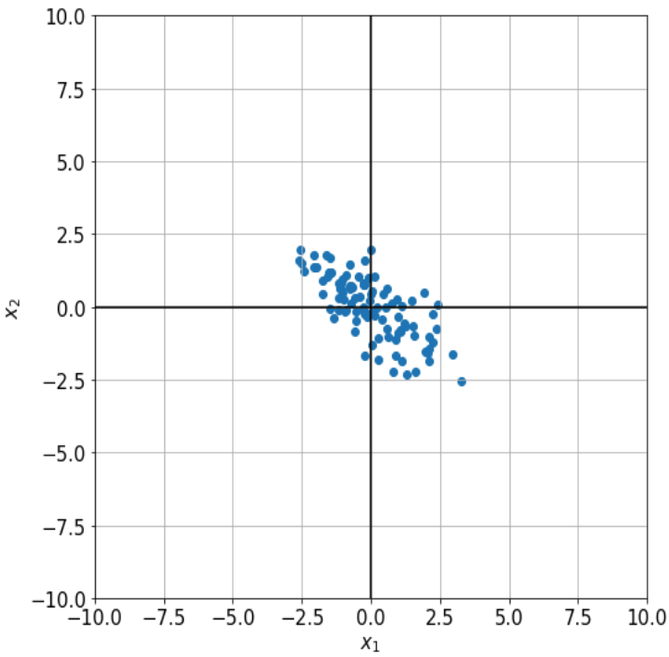
3. Covariance matrix

$\boxed{X}$

$C = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^T$

$\boxed{Av = \lambda v}$

$C = \frac{1}{n} X X^T$



$$\mathbf{C} = \begin{bmatrix} 1 & -0.9 \\ -0.9 & 1 \end{bmatrix}$$

(i) $C^T = C$

(ii) P.S.D ✓

(iii) Orthogonal / orthonormal

<u>Shape</u>

$$\mathbf{C} \in \mathbb{R}^{d \times d}$$

Dot product

Outer product

<u>Outer-product form</u>

eq.

$$\mathbf{C} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^T$$

<u>Matrix-form</u>

$$\mathbf{C} = \frac{1}{n}\mathbf{X}\mathbf{X}^T$$

<u>Scalar form</u>

$$C_{pq} = \frac{1}{n}\sum x_{ip} x_{iq}$$

C_{pq} captures the covariance between the pth feature and the qth feature.

Outer product:

$x \in \mathbb{R}^d \qquad y \in \mathbb{R}^p$

$$\boxed{xy^T}$$

$$A = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix}_{d\times 1} \begin{bmatrix} y_1 & \cdots & y_p \end{bmatrix}_{1\times p} = \begin{bmatrix} & \\ & \end{bmatrix}_{d\times p}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1\times 4 & 1\times 5 & 1\times 6 \\ 2\times 4 & 2\times 5 & 2\times 6 \\ 3\times 4 & 3\times 5 & 3\times 6 \end{bmatrix}$$

As a special case:

$$C_{pp} = \frac{1}{n}\sum x_{ip}^2$$

C_{pp} captures the variance of the pth feature.

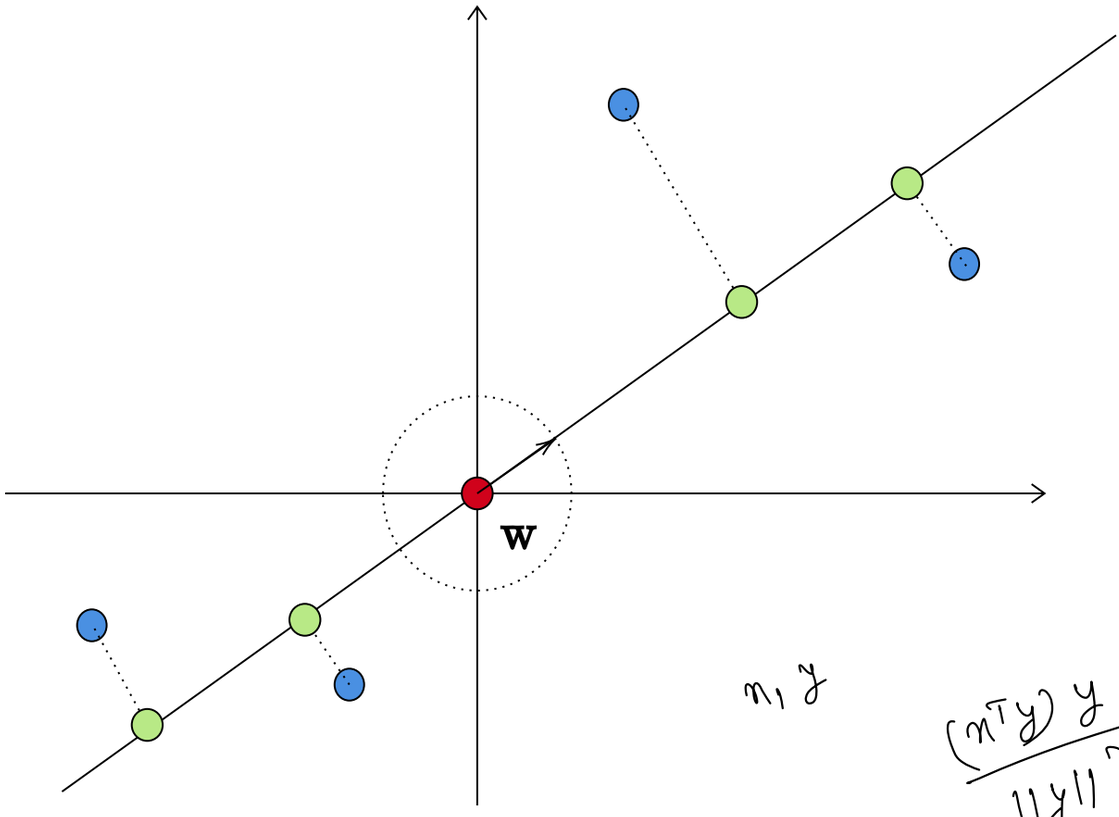
<u>Properties</u>

- $\mathbf{C}^T = \mathbf{C}$
- All eigenvalues of $\mathbf{C}$ are non-negative.
  - $\lambda_1 \geqslant \cdots \geqslant \lambda_d \geqslant 0$
- There is an orthonormal basis for $\mathbb{R}^d$ made up of eigenvectors of $\mathbf{C}$
  - $\{\mathbf{w}_1, \cdots, \mathbf{w}_d\}$
  - This comes from the spectral theorem.

Note: If $\mathbf{C}$ is a square matrix, then $(\lambda, \mathbf{w})$ is said to be an eigenvalue-eigenvector pair if $\mathbf{C}\mathbf{w} = \lambda\mathbf{w}$. Note that $\mathbf{w} \neq \mathbf{0}$ for it to be an eigenvector.

Remark: $\mathbf{w}_i$ will always represent a unit-norm vector in the rest of the document.

4. Optimization problem



x, y

$$\frac{(x^T y)\, y}{\|y\|^2}$$

<u>Minimizing the reconstruction error</u>

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x^2 + 4xy + y^2$$

$$(2x + 4y,\ 2y + 4x)$$

$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} ||\mathbf{x}_i - \left(\mathbf{x}_i^T \mathbf{w}\right)\mathbf{w}||^2$$

$$\boxed{x^T A x}$$

$$f(x)$$

<u>Maximizing the variance</u>

$$g(w) = w^T w = 1$$

$$= w^T w - 1 = 0$$

$$g(x) = 0$$

$$\max_{\mathbf{w}} \mathbf{w}^T \mathbf{C} \mathbf{w}$$

$$L(x, \lambda) = f(x) - \lambda g(x)$$

$$w^T C w - \lambda(w^T w - 1)$$

Both forms are equivalent to each other.

5. Principal components

$$\frac{\partial L}{\partial w} = 2Cw - 2\lambda w = 0$$

$$Cw = \lambda w$$

$$\boxed{ax^2}$$

$$\boxed{x^T A x} \rightarrow (A + A^T)\, x$$

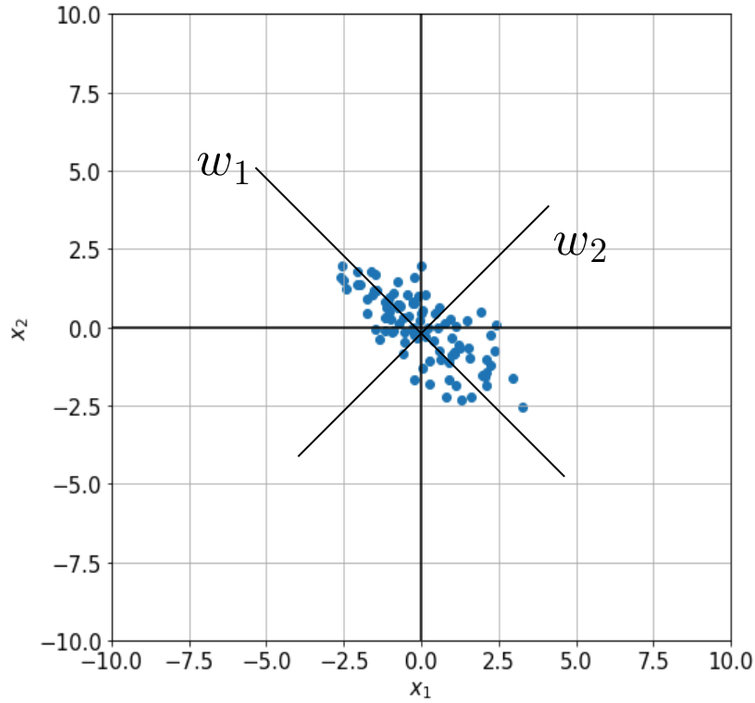
$$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

$$w^T w = w_1^2 + w_2^2 + w_3^2$$

$$= (2w_1, 2w_2, 2w_3)$$

$$= 2(w_1, w_2, w_3)$$

$$= 2w$$



Let $(\lambda_1, \mathbf{w}_1), \cdots, (\lambda_d, \mathbf{w}_d)$ be the eigen-pairs of $\mathbf{C}$, where $\lambda_1 \geqslant \cdots \geqslant \lambda_d$ and $\{\mathbf{w}_1, \cdots, \mathbf{w}_d\}$ is an orthonormal basis for $\mathbb{R}^d$. $\mathbf{w}_i$ is termed the ith principal component of $\mathbf{C}$. To be more precise:

$$\mathbf{C}\mathbf{w}_i = \lambda_i \mathbf{w}_i$$

$$\mathbf{w}_i^T \mathbf{w}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$(AB)^T = B^T A^T$

$$\lambda_1 = \max_{\mathbf{w}} \quad \mathbf{w}^T \mathbf{C} \mathbf{w}$$

$(x^T w)^T (x^T w)$

$(x^T w)$

$w^T \boxed{x \, x^T} w$

$w^T C w$

$$\mathbf{w}_1 = \arg\max_{\mathbf{w}} \quad \mathbf{w}^T \mathbf{C} \mathbf{w}$$

$$\boxed{\mathbf{w}_1^T \mathbf{C} \mathbf{w}_1 = \lambda_1}$$

Threshold $= 0.95$

$$\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{d} \lambda_i}$$

$n = 100$

$d = 5$

$x_i \in \mathbb{R}^5$

$C_{d \times d}$

$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \cdots \lambda_5}$$

$$\theta = \frac{\text{Var. along } 1^{st} \text{ PC}}{\text{sum of var. along } 5 \text{ PCs}}$$

find minimum value of θ than can be possible

$$\theta = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5}$$

$\lambda_1 \geq \lambda_2 \geq \cdots \lambda_5$

when $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5$

$$\left.\begin{array}{l} \lambda_1 = 5 \\ \lambda_2 = 5 \\ \vdots \quad 5 \\ \vdots \quad = \vdots \\ \lambda_5 = 5 \end{array}\right\} \Rightarrow 25$$

15

$= 9$

$$\theta = \frac{\lambda_1}{\lambda_1 + \lambda_1 + \cdots \lambda_1}$$

$$= \frac{\lambda_1}{5\lambda_1}$$

$$= \frac{1}{5} = 0.2$$

6. Projections

$x_1 \in \mathbb{R}^d$

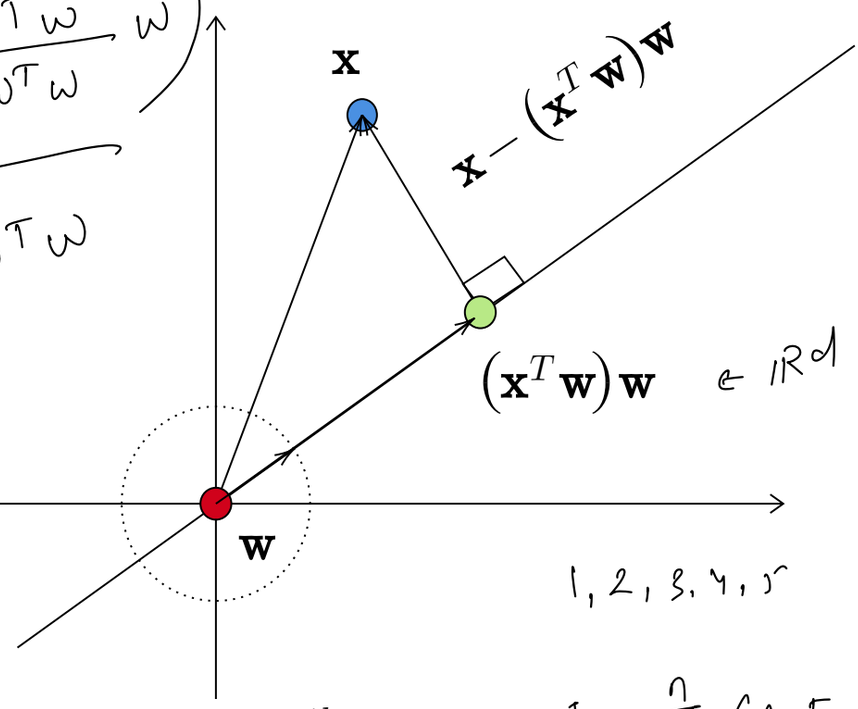
$w \in \mathbb{R}^d$

$$\sqrt{\left(\frac{x^T w}{w^T w} w\right)^T \left(\frac{x^T w}{w^T w} w\right)}$$

$$= \sqrt{\frac{(x^T w)^2}{(w^T w)^2} w^T w}$$

$$= \frac{x^T w}{\sqrt{w^T w}}$$

$$= x^T w$$



1, 2, 3, 4, 5

(Vector) Projection of $\mathbf{x}_i$ onto the jth PC

$$\frac{1}{n} \sum_{i=1}^{n} \left((x^T w) w\right)^2$$

$$\left(\mathbf{x}_i^T \mathbf{w}_j\right)\mathbf{w}_j$$

Scalar projection of $\mathbf{x}_i$ onto the jth PC (or) coordinate of the data-point along this direction:

$$\mathbf{x}_i^T \mathbf{w}_j$$

The projection of a data-point $\mathbf{x}_i$ onto the top k principal components:

$$\mathbf{x}_i' = \left(\mathbf{x}_i^T \mathbf{w}_1\right)\mathbf{w}_1 + \cdots + \left(\mathbf{x}_i^T \mathbf{w}_k\right)\mathbf{w}_k$$

To represent the reconstruction and scalar projections in matrix form:

$$\mathbf{W} \in \mathbb{R}^{d \times k}$$

$$u = u_\perp + w_p$$

$$u = u_\perp + cw$$

$$u^T w = (u_\perp + cw)^T w$$

$$w \qquad = u^T w = u_\perp^T w + c w^T w$$

$$= u^T w = c w^T w$$

$$c = \frac{u^T w}{w^T w}$$

$$V = \{ (x, y, z) \mid x + y + z = 0 \}$$

$\in \mathbb{R}^3$

$$x_i \in \mathbb{R}^{100}$$

$$(x_i^T w_1) w_1 + (x_i^T w_2) w_2 + \cdots$$

$$\begin{bmatrix} x_i^T w_1 \\ x_i^T w_2 \end{bmatrix}$$

$$0 = x_i - (x_i^T w_1) w_1 - (x_i^T w_2) w_2$$

$$x_i = (x_i^T w) w_1 + (x_i^T w_2) w_2$$

$$\mathbf{W} = \begin{bmatrix} | & & | \\ \mathbf{w}_1 & \cdots & \mathbf{w}_k \\ | & & | \end{bmatrix}$$

<u>Scalar projections</u>

$$\mathbf{X}' \in \mathbb{R}^{k \times n}$$

$$\mathbf{X}' = \begin{bmatrix} \mathbf{x}_1^T \mathbf{w}_1 & & \mathbf{x}_n^T \mathbf{w}_k \\ | & \cdots & | \\ \mathbf{x}_1^T \mathbf{w}_k & & \mathbf{x}_n^T \mathbf{w}_k \end{bmatrix}$$

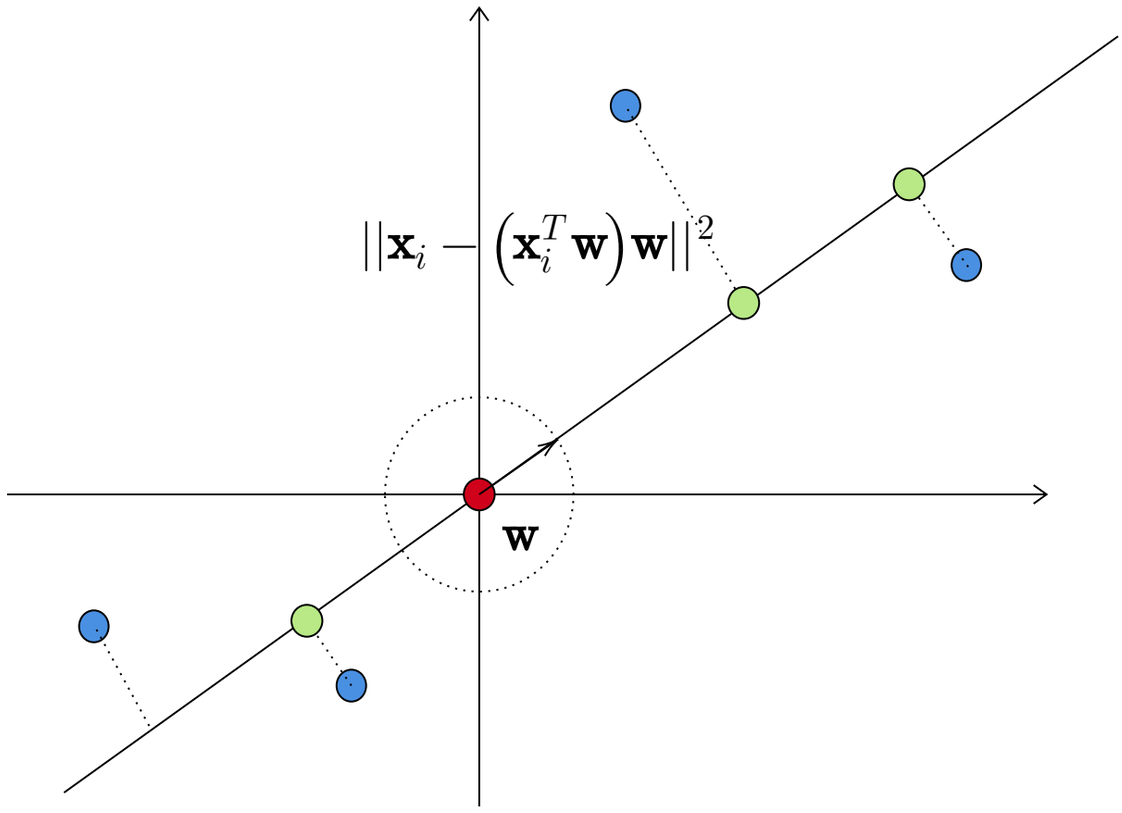
$$\mathbf{X}' = \mathbf{W}^T \mathbf{X}$$

<u>Reconstruction</u>

$$\mathbf{X}' \in \mathbb{R}^{d \times n}$$

$$\mathbf{X}' = \mathbf{W}\mathbf{W}^T \mathbf{X}$$

7. Reconstruction error revisited (for k directions)



$$\frac{1}{n}\sum_{i=1}^{n} ||\mathbf{x}_i - \mathbf{x}_i'||^2$$

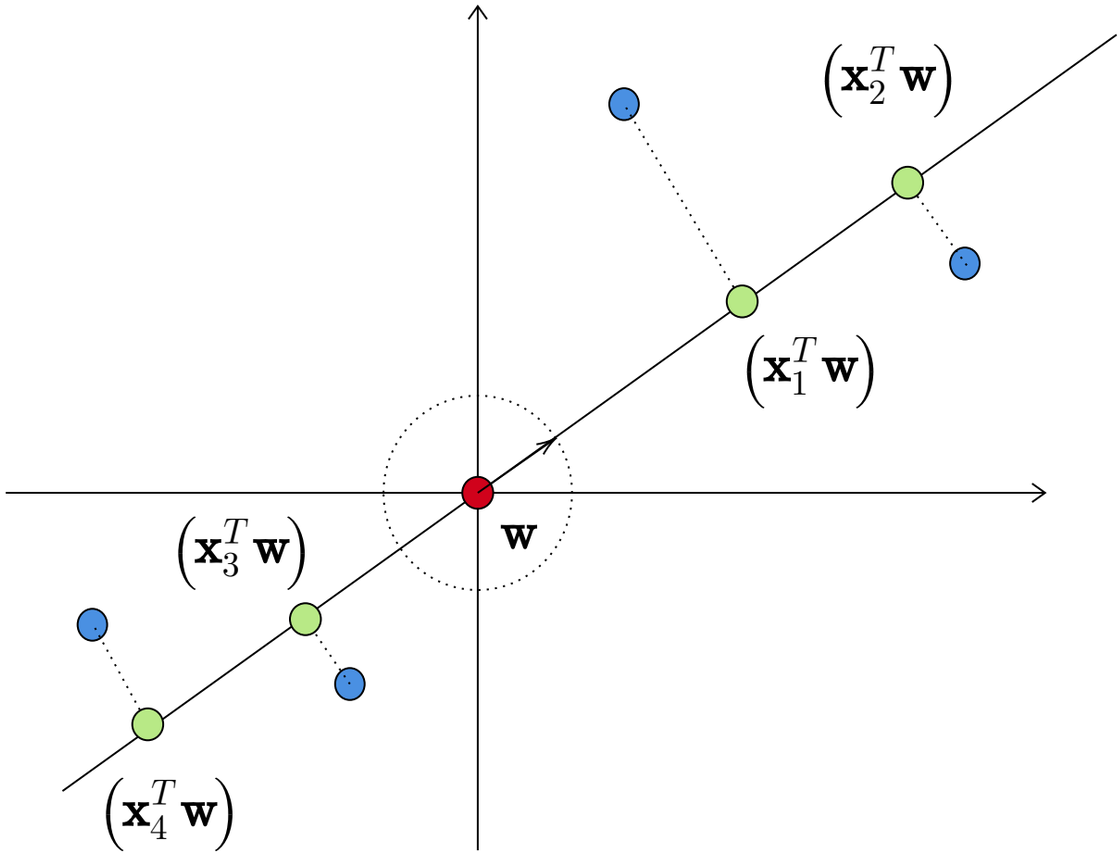
$$\frac{1}{n}\sum_{i=1}^{n} \left|\left|\mathbf{x}_i - \sum_{j=1}^{k} \left(\mathbf{x}_i^T \mathbf{w}_j\right)\mathbf{w}_j\right|\right|^2$$

8. Variance captured

Total variance:

$$\lambda_1 + \cdots + \lambda_d$$

Variance along a given direction $\mathbf{w}$ (unit vector):



$$\frac{1}{n}\sum_{i=1}^{n}\left(\mathbf{x}_i^T\mathbf{w}\right)^2$$

$$\mathbf{w}^T\mathbf{C}\mathbf{w}$$

Proportion of variance captured by top k PCs:

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_d}$$

Heuristic to choose the value of k: smallest value that captures 95% of the variance in the dataset.

9. Compression

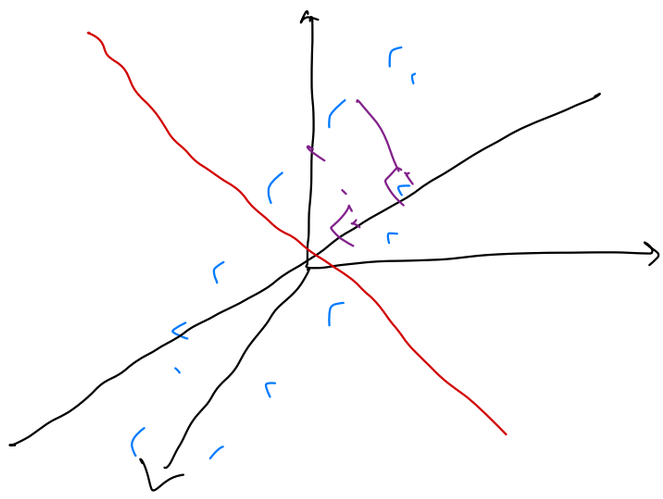
Reconstruction

$$\frac{nk + dk}{dn} = \frac{k(d+n)}{dn}$$

Retaining only scalar projections

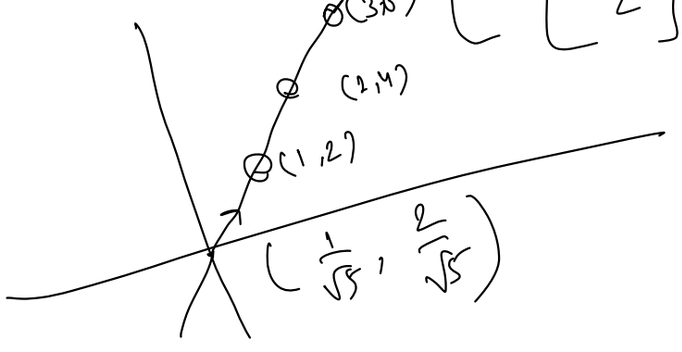
$$\frac{kn}{dn} = \frac{k}{d}$$

$$\boxed{\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3}}$$



$$\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{n} \lambda_i}$$

$$X = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix} \right\}$$



$$\frac{nk + dk}{dn} = \frac{k(d+n)}{dn}$$

Retaining only scalar projections

$$\frac{kn}{dn} = \frac{k}{d}$$

$$S = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 8 \end{bmatrix}, \begin{bmatrix} 1/5 \\ 4/15 \end{bmatrix} \right\}$$

(i) $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

(ii) $\begin{bmatrix} 1/15 \\ 4/15 \end{bmatrix}$

(iii) $\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$